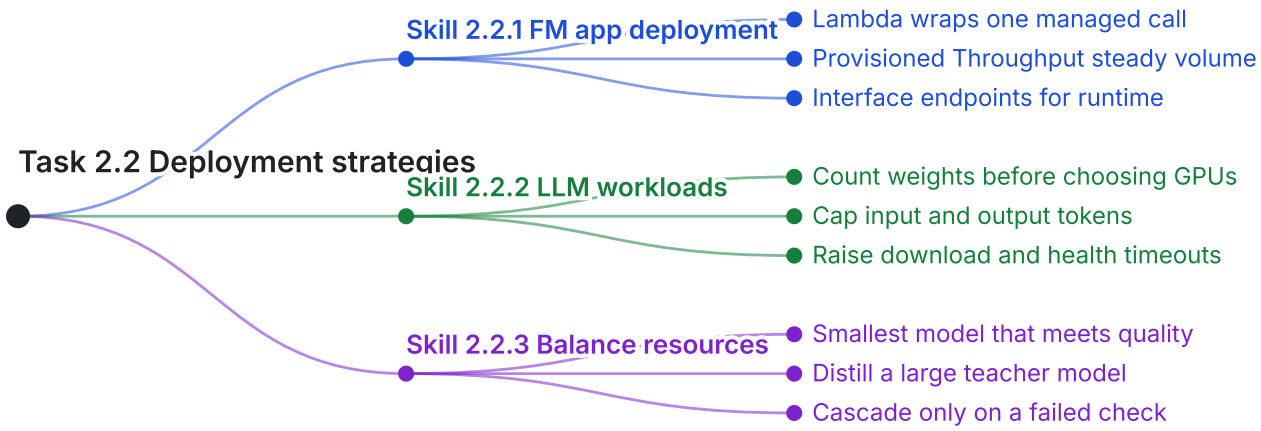
Memory sheet: Task 2.2 Model deployment strategies

Domain 2, weight 26 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Each request arrives with no capacity planning	On-demand Bedrock invocation	The model is customized and has no deployment for it
Guaranteed throughput or model units are named	Provisioned Throughput on the model itself	The target is a cross-Region inference profile
The caller waits for each answer on the same call	Real-time endpoint for a model that fits	The payload is large or processing takes minutes
The payload sits in S3 and the result can wait	Asynchronous endpoint	A user is waiting for a live chat answer
The endpoint sits idle most hours and cold starts are fine	Serverless endpoint	The model needs a GPU or a VPC configuration
One full dataset runs on a schedule with no endpoint	Batch transform job	Each request needs its own answer
Several models in one family must balance cost and quality	Intelligent prompt routing	Routing depends on tenant rules or your own review scores
The team owns the weights and wants the Bedrock API	Custom Model Import	New training is required, since import does not train

Etch lines

1. A GPU or VPC need rules out a serverless endpoint.
2. Batch transform writes one output file per input file with an .out extension.
3. A NAT gateway does not help a container under network isolation.
4. Set an output cap on every route, then record input and output token counts.

5. Send each request to the smallest model that meets the target.
6. Distillation does not make the student better than the teacher.

Flash questions

- Q: Which endpoint type fits a payload in S3 when the caller reads the result later? A: An asynchronous endpoint, which queues requests and writes each result to S3.
- Q: Why can a serverless endpoint not host a model that needs a GPU? A: Serverless endpoints do not support GPUs, so a real-time endpoint is needed.
- Q: What commitment term lets you delete Provisioned Throughput whenever you choose? A: No commitment, though billing continues until you delete it.
- Q: Which interface endpoint carries Bedrock inference calls over a private path? A: The bedrock-runtime interface endpoint, since the control plane endpoint does not carry inference.
- Q: What lets a model container with no internet access read its weights from S3? A: An S3 gateway endpoint attached to the route tables of its subnets.
- Q: How do you estimate weight memory for an LLM before choosing GPUs? A: Multiply the parameter count by the bytes per parameter, then add room for buffers.
- Q: What does quantization change, and what does it leave alone? A: It lowers weight precision to cut memory, and it leaves the architecture and training data unchanged.
- Q: Why can one output limit shared by every route cause a problem? A: Routes differ in size, so a shared limit can leave a short-answer route too loose.

Source: Task 2.2 skill map, Skills 2.2.1 to 2.2.3, The deciding constraint, and Private network placement for hybrid endpoints. AWS docs and the study guide sections above, no practice-test text.